INCH-POUND

MIL-DTL-17/90B
w/AMENDMENT 1
06 February 2026
SUPERSEDING
MIL-DTL-17/90B
19 June 2015

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, COAXIAL, 93 OHMS, M17/90-RG71

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

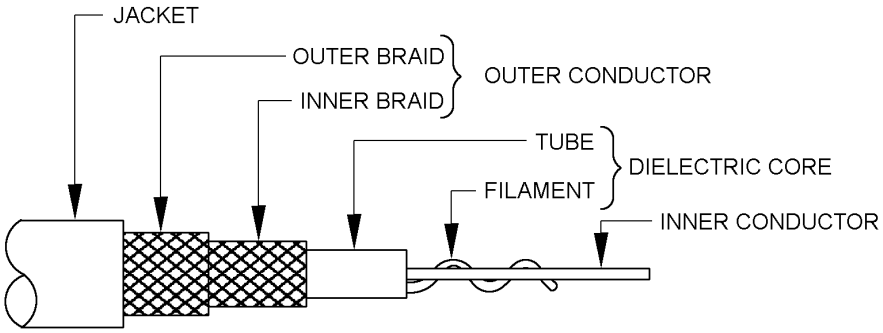


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductors	Solid, copper-covered, steel wire. Diameter: .0253 ± .0010.
Dielectric core	Type A-3: Air-spaced polyethylene. A monofilament thread, .035 inch approximate diameter with a lay of 1/2 inch approximate, under an extruded tube. Diameter: .146 inch ± .005. Alternate: A continuous tube, .003 inch thick maximum, under two continuous spiral fins, with a lay of 1-1/4 inches approximate, under an extruded tube.
Outer conductor	Double braid of copper wire. Diameter: .208 inch, maximum.
	Alternate
Inner braid	Wire gauge: AWG #34 AWG #36 Wire coating: Bare Tinned Coverage: 94.3% nominal 95.2% nominal Carriers: 16 24 Ends: 7 6 Picks/inch: 8.2 ± 10% 12.0 ± 10%
Outer braid	Wire gauge: AWG #36 AWG #36 Wire coating: Tinned Tinned Coverage: 93.8% nominal 94.6% nominal Carriers: 16 24 Ends: 9 6 Picks/inch: 10.6 ± 10% 16.0 ± 10%
Jacket	Type IIIa. Diameter: .245 inch ± .005.

AMSC N/A
FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 84 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +85°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 4.4 ohms per 100 feet.
Elongation: 1 percent, minimum.
Tensile strength: 110 $klb_f/inch^2$, minimum.
Engineering notes: This cable useful in low temperature applications (see connector series TNC or BNC per MIL-PRF-39012).

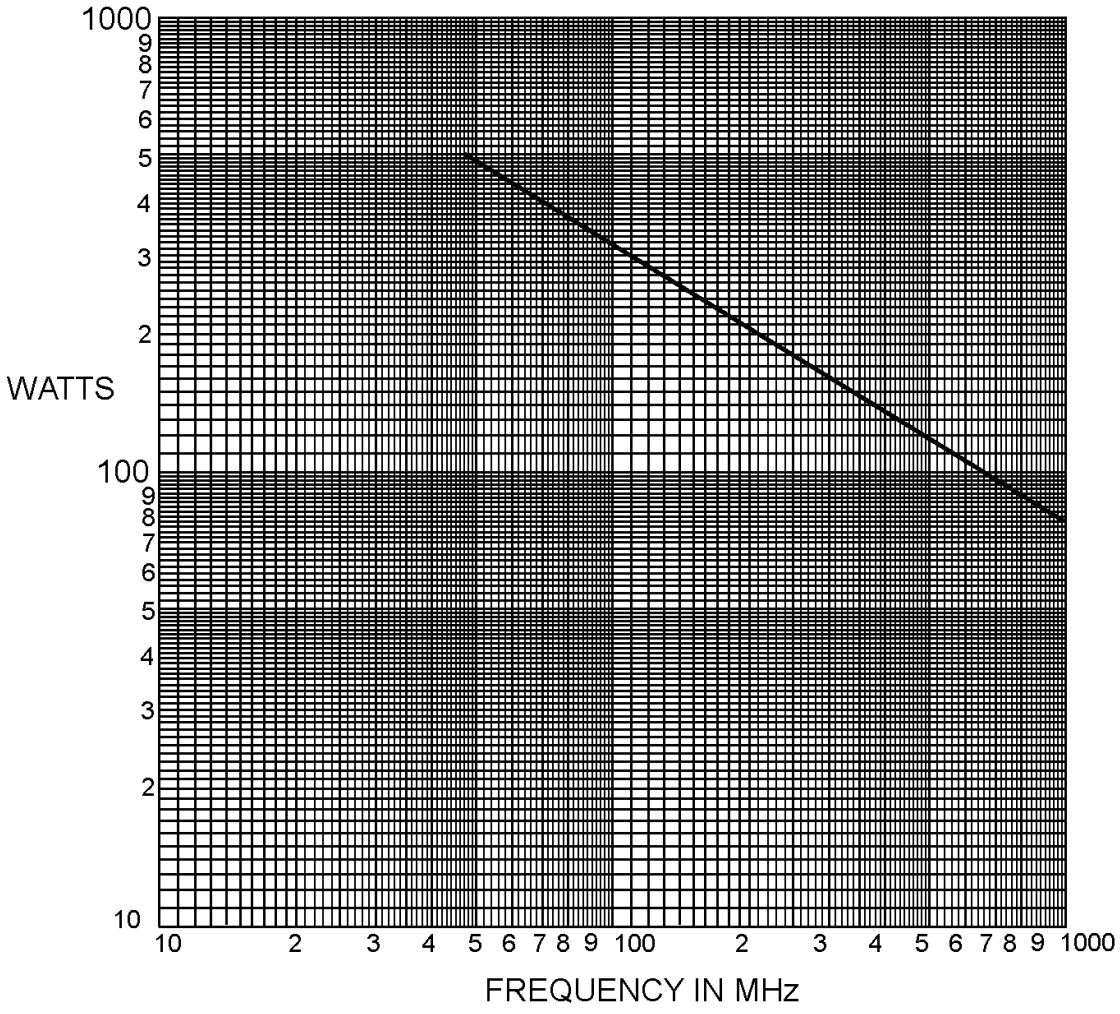
REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors: Not applicable.
Aging stability: +90° ± 2°C.
Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Cold bend: -55 ± 2°C.
Dimensional stability: Not applicable.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 5 pounds per 100 feet, maximum.

Electrical:
Test frequency: 50 MHz to 1 GHz.
Spark test: 5,000 V rms, +25%, -0%.
Voltage withstanding: 3,000 Vrms, minimum.
Insulation resistance: Not applicable.
Corona extinction voltage: Not applicable.
Characteristic impedance: 93 ohms ± 5.
Attenuation: 8.0 dB per 100 feet at 400 MHz.
Structural return loss: Not applicable.
Capacitance: 14.5 pF per foot, maximum.
Capacitance stability: ± 1.5 percent.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

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Power	
MHz	Watts
50	480
100	310
200	200
400	135
1000	77

FIGURE 2. Power rating.

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Part or Identifying Number (PIN): M17/90-RG71.
Marking: Applicable
Marking Durability: 125 cycles (250 strokes), 1 pound
Workmanship: Applicable

Supersession data: See table II.

Qualification: Shall consist of the inspections specified in table III.

TABLE II. Cross reference of part number.

PIN	Superseded PIN or type designation
M17/90-RG71	RG71B/U

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TABLE III. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	3.6.2	4.8.1.2	Entire Sample		
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	N/A				
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	N/A				
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	3.7.10	4.8.11	1	Per Table VIII	C
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	3.7.15	4.8.16	4	Per Table VIII	C

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TABLE III. Qualification and Group A, B, and C Table – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	N/A				
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion	N/A				
	Smoke index	N/A				
	Toxicity index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals (aged)	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2026-002)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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